11

Electric Circuits

11.8

Resistor-Capacitor (RC) Circuits

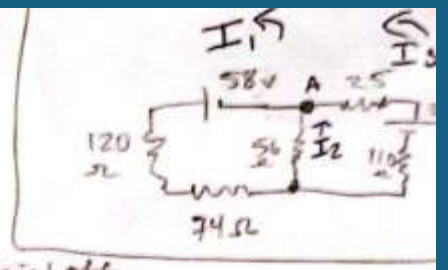
11.8.1

Equivalent Capacitance in Series and Parallel

-Individual Capacitor Potential Difference (AKA Voltage) and Plate Charges

30 At Point A, a junction Is formed where

- $I_1 = I_2 + I_3$ combine according to kirchoff



Left Loop (Counterclockwise)

$$58v - (I_1)(120\Omega) - I_1(74\Omega) - I_2(56\Omega) = 0$$

$$58v - [(I_1)(120+74)] - [(I_2)56] = 0$$

- $58v = 194 I_1 + 56 I_2$

Right Loop (Counter Clockwise)

$$3v - I_3(25\Omega) + I_2(56\Omega) - I_3(110) = 0$$

$$3v - [(I_3)(135)] + [I_2(56)] = 0$$

- $3v = 135 I_3 - 56 I_2 \rightarrow 56 I_2 = 135 I_3 - 3V$

If $V = 5.8$

① $5.8 = 194 I_1 + 56 I_2$

② $3 = -56 I_2 + 135 I_3$

③ $I_1 = I_2 + I_3$

Insert $I_1 = I_2 + I_3$

① $5.8 = 250 I_2 + 194$

② $3 = -56 I_2 + 135 I_3$

① times 56 $\rightarrow$ $324.8 = 14000 I_2 + 10864 I_3$

② times 250 $\rightarrow$ $750 = -14000 I_2 + 33750 I_3$

$1074.8 = 44614 I_3$

$I_3 = .02409A$

$3 = -56 I_2 + 135(.02409)$

$-.252297A = -56 I_2$

$I_2 = .004505$ Amps

$I_1 = I_2 + I_3$

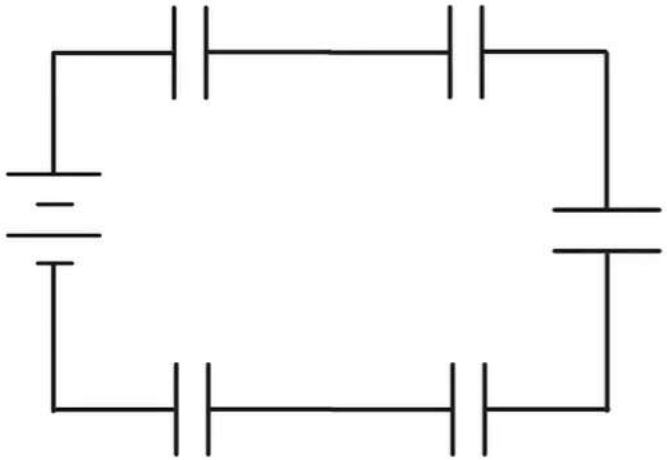
$I_1 = .02409 + .004505$

$I_1 = .028595$ Amps

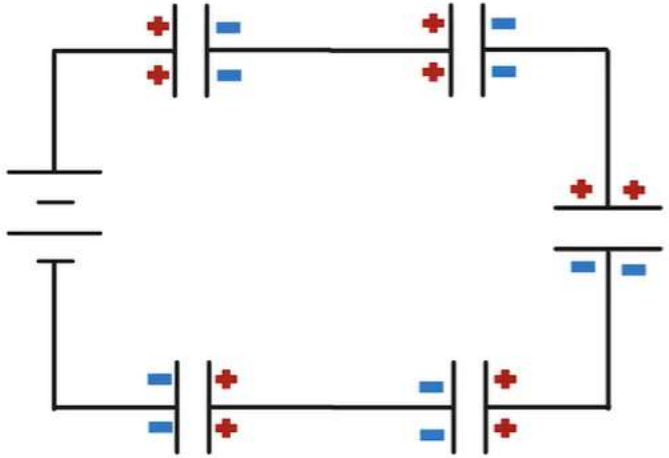
Series vs. Parallel

Resistors in Series	Resistors in Parallel
• Current only has **one** path to travel. • Current is the same for all resistors on the path. • More obstacles on the path = **more** resistance.	• Current has **multiple** paths it could travel. • Potential difference is the same across all resistors in the combo. • More paths = **less** resistance.
$R_{eq,series} = \sum_n R_n = R_1 + R_2 + \cdots + R_n$	$\frac{1}{R_{eq,parallel}} = \sum_n \frac{1}{R_n} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$

Capacitors in Series



Capacitors in Series



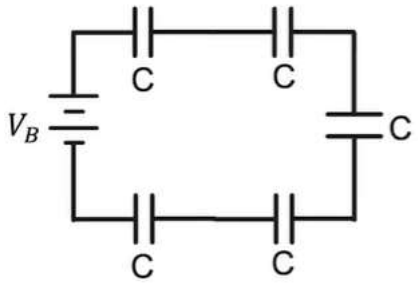
Current only has one path to travel

Charge is the same for all capacitors on the path

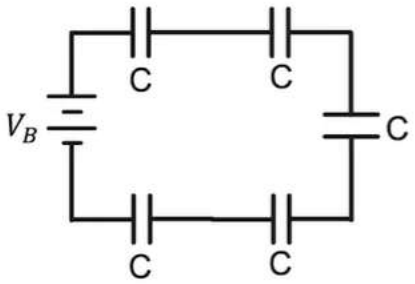
More capacitors on the path = LESS Capacitance

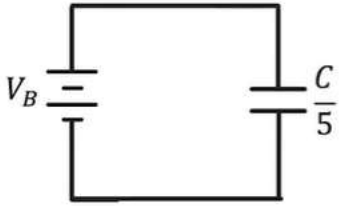
$$\frac{1}{C_{eq,series}} = \sum_n \frac{1}{C_n} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

Capacitors in Series



Capacitors in Series





$$\frac{1}{C_{eq,series}} = \sum_n \frac{1}{C_n} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C} = \frac{5}{C}$$

$$C_{eq} = \frac{C}{5}$$

Entire Circuit

$Q = C\Delta V$

$$Q = \frac{CV_B}{5}$$

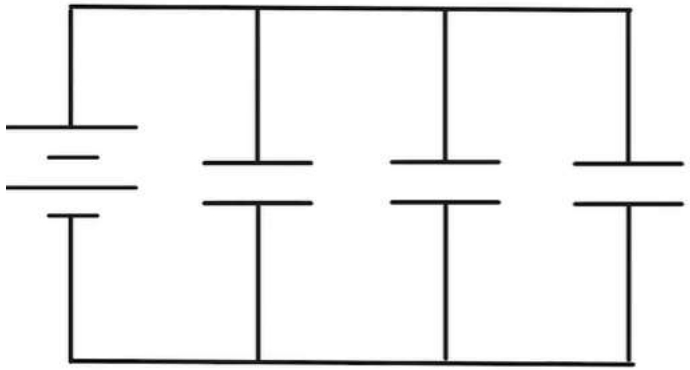
Individual Capacitors

$Q = C\Delta V$

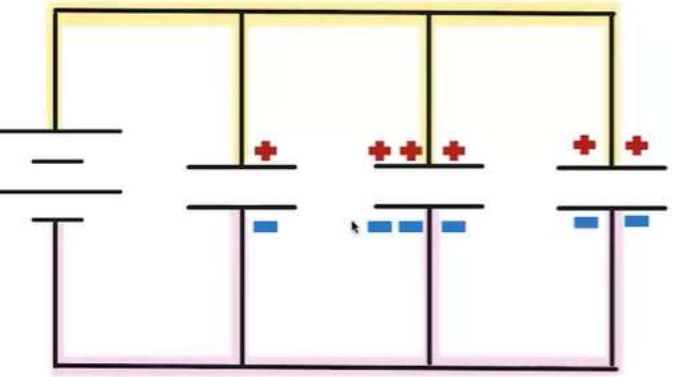
$$V_C = \frac{Q}{C}$$

$$V_C = \frac{V_B}{5}$$

Capacitors in Parallel



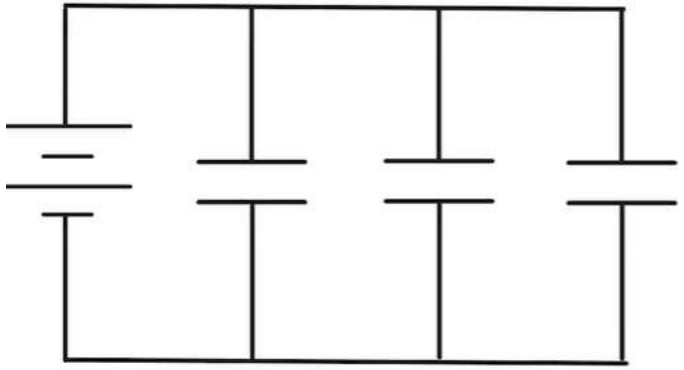
Capacitors in Parallel



Current has multiple paths it could travel- Different charge!

Potential difference is the same across all capacitors

Capacitors in Parallel



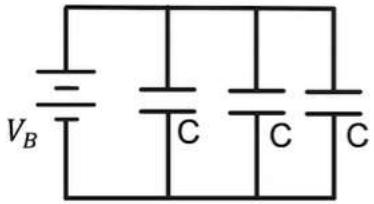
Current has multiple paths it could travel- Different charge!

Potential difference is the same across all capacitors

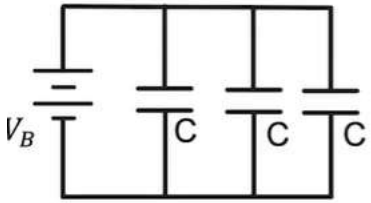
More paths = More Capacitance

$$C_{eq,parallel} = \sum C_n = C_1 + C_2 + \cdots + C_n$$

Capacitors in Parallel



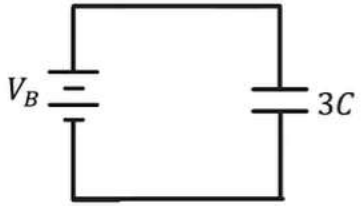
Capacitors in Parallel



$$C_{eq,parallel} = \sum_n C_n = C_1 + C_2 + \cdots + C_n$$

$$C_{eq} = C + C + C = 3C$$

$$\boxed{C_{eq} = 3C}$$



Entire Circuit

$Q = C\Delta V$

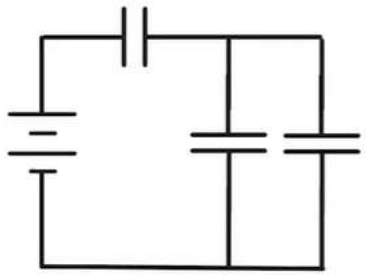
$$\boxed{Q = 3CV_B}$$

Individual Capacitors

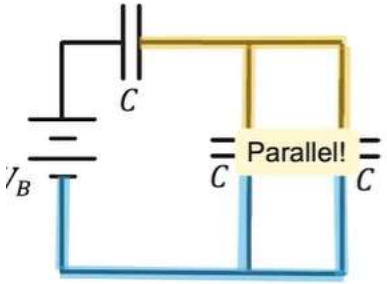
$Q = C\Delta V$

$$\boxed{Q = CV_B}$$

Capacitors in Series and Parallel Combination

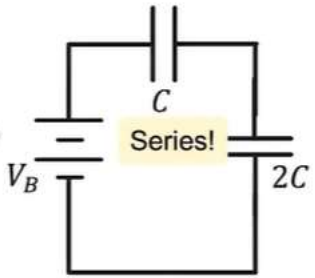


Capacitors in Series and Parallel Combination



$$C_{eq} = C + C$$

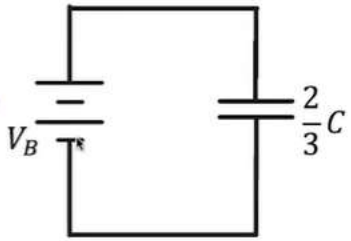
$$C_{eq} = 2C$$



$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{2C}$$

$$\frac{1}{C_{eq}} = \frac{3}{2C}$$

$$C_{eq} = \frac{2C}{3}$$

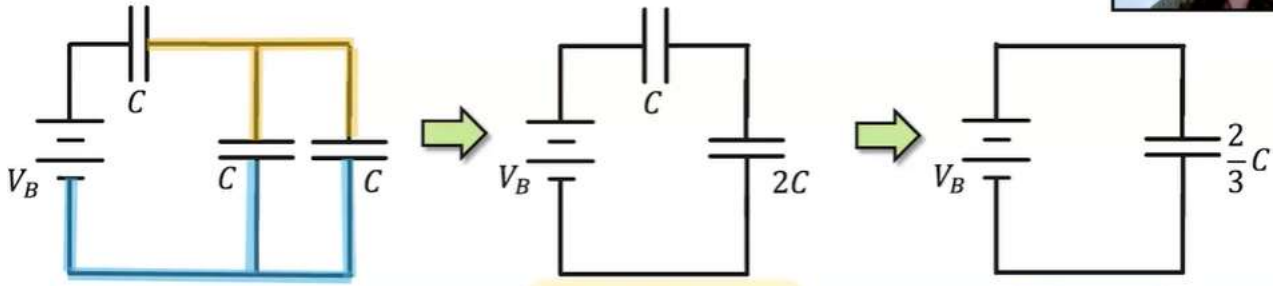


$$C_{eq} = \frac{2}{3}C$$

$$Q = C(\Delta V)$$

$$Q = \frac{2CV_B}{3}$$

Capacitors in Series and Parallel Combination



Parallel! They have the same voltage as their combo

Series! They have the same charge as the whole circuit

$$C_{eq} = \frac{2}{3}C$$

$$Q = C(\Delta V)$$

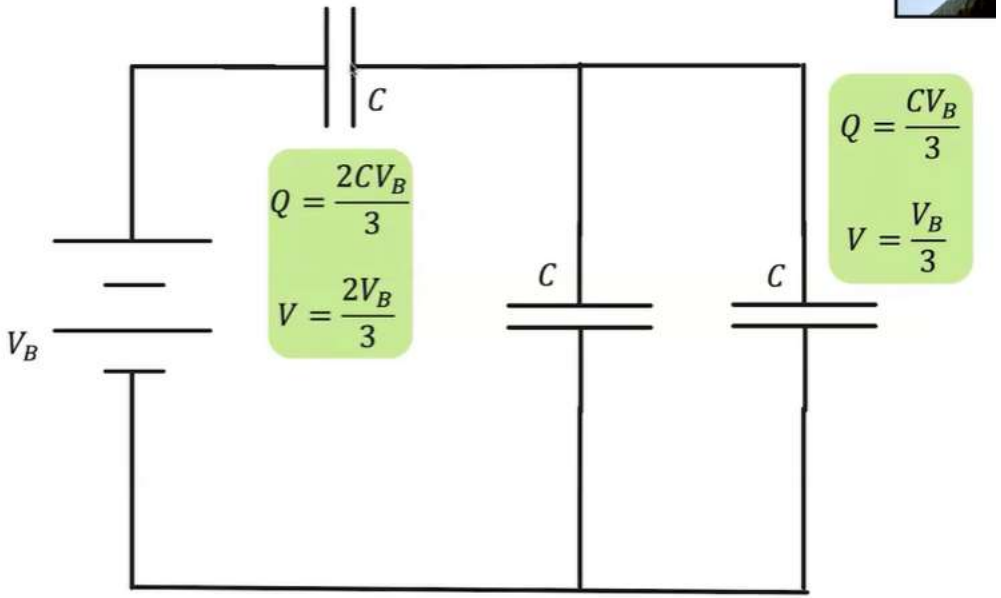
$$Q = \frac{2CV_B}{3}$$

$$V_C = \frac{2CV_B}{3} \cdot \frac{1}{C} = \frac{2V_B}{3}$$

$$V_{2C} = \frac{2CV_B}{3} \cdot \frac{1}{2C} = \frac{V_B}{3}$$

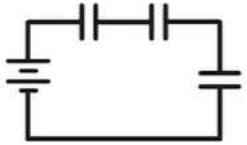
$$Q = \frac{CV_B}{3}$$

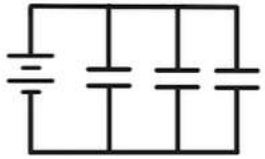
Capacitors in Series and Parallel Combination



Takeaways

Capacitors in Series	Capacitors in Parallel
• Current only has **one** path to travel • **Current** is the same for all capacitors on the path- so the CHARGE on the capacitors is also the same • Equivalent capacitance is **less** than smallest capacitor	• Current has **multiple** paths it could travel • **Potential difference** is the same across all capacitors in the combo • Equivalent capacitance is the **sum** of all capacitors in parallel





Takeaways

- More capacitors in **series** = **less** equivalent capacitance
- More capacitors in **parallel** = **more** equivalent capacitance
- Equivalent capacitances help you find plate charges and voltages across individual capacitors

$$\frac{1}{C_{eq,series}} = \sum_n \frac{1}{C_n} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

$$C_{eq,parallel} = \sum_n C_n = C_1 + C_2 + \cdots + C_n$$

11.8.2

Kirchhoff’s Loop Rule For Current and Voltage Drops in RC Circuits

Kirchhoff's Loop Rule:

A complete loop around a circuit has a potential difference of ZERO

$$\sum \Delta V = 0$$

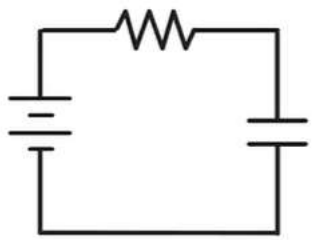
Voltage across resistor:

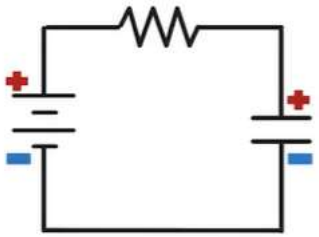
$$\Delta V = IR$$

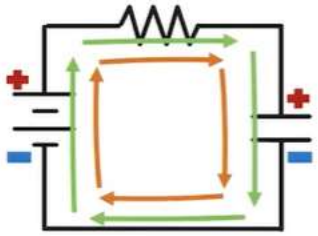
Voltage across capacitor:

$$\Delta V = \frac{Q}{C}$$

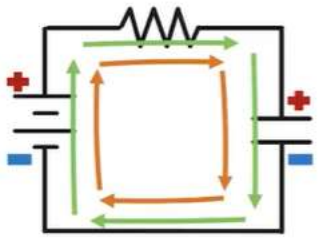
Simple RC Circuit







Simple RC Circuit



$$+V_{battery} - V_{resistor} - V_{capacitor} = 0$$

$$+V_b - IR - V_c = 0$$

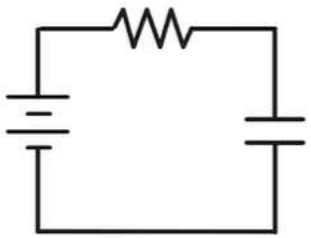
Remember that:

$Q = CV_c$ and $I = \frac{dQ}{dt} \rightarrow C\frac{dV_c}{dt}$

$$+V_b - RC\frac{dV_c}{dt} - V_c = 0$$

DiffEQ Time!

Simple RC Circuit



$$+V_b - RC\frac{dV_c}{dt} - V_c = 0$$

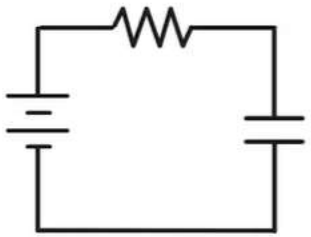
$$-RC\frac{dV_c}{dt} = V_c - V_b$$

$$-RCdV_c = (V_c - V_b)dt$$

$$\frac{dV_c}{V_c - V_b} = -\frac{dt}{RC}$$

$$\int_{V_{c0}}^{V_c} \frac{dV_c}{V_c - V_b} = \int_0^t -\frac{dt}{RC}$$

Simple RC Circuit



$$\int \frac{dx}{x+a} = \ln|x+a|$$

This integral is on the equation sheet!

$$\int_{V_{c0}}^{V_c} \frac{dV_c}{V_c - V_b} = \int_0^t -\frac{dt}{RC}$$

$$\ln|V_c - V_b| \Big|_0^{V_c(t)} = -\frac{t}{RC}\Big|_0^t$$

Note: the battery will be removed, so the battery voltage term will NOT be included.

Vc = Vci = Vo = 0

Also, For a

DISCHARGING Capacitor

The LHS bounds will be reversed

AND

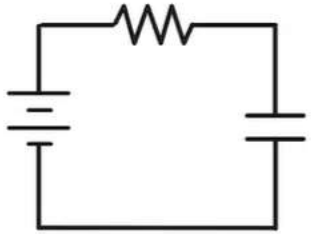
Vb = 0 since the battery is disconnected

$$\ln|V_c(t) - V_b| - \ln|0 - V_b| = -\frac{t}{RC}$$

$$\ln\left|\frac{V_c(t) - V_b}{-V_b}\right| = -\frac{t}{RC}$$

$$\ln\left|-\frac{V_c(t)}{V_b} + 1\right| = -\frac{t}{RC}$$

Simple RC Circuit



$$\ln\left|-\frac{V_c(t)}{V_b}+1\right| = -\frac{t}{RC}$$

$$-\frac{V_c(t)}{V_b}+1 = e^{-\frac{t}{RC}}$$

$$\frac{V_c(t)}{V_b} = 1 - e^{-\frac{t}{RC}}$$

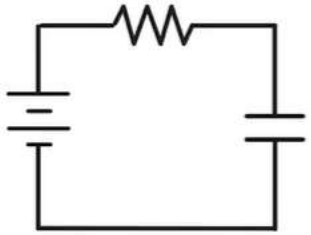
$$Q = C(\Delta V)$$

$V_c(t) = V_b\left(1 - e^{-t/RC}\right)$
Voltage across a **CHARGING** capacitor

$Q(t) = CV_b\left(1 - e^{-t/RC}\right)$
Charge on a **CHARGING** capacitor

Simple RC Circuit

To get current and voltage for the resistor, remember that $I = \frac{dQ}{dt}$ and from $V_b - V_R - V_c = 0$, $V_R = V_b - V_c$



$$V_R(t) = V_b e^{-t/RC}$$

Voltage across the resistor when **CHARGING** capacitor

$$\mathrm{I}(t) = \frac{V_b}{R} e^{-t/RC}$$

Current through resistor when **CHARGING** capacitor

Takeaways

$V_c(t) = V_b\left(1 - e^{-t/RC}\right)$
Voltage across a **CHARGING** capacitor

$Q(t) = CV_b\left(1 - e^{-t/RC}\right)$
Charge on a **CHARGING** capacitor

$V_R(t) = V_b e^{-t/RC}$
Voltage across the resistor when **CHARGING** capacitor

$\mathrm{I}(t) = \frac{V_b}{R} e^{-t/RC}$
Current through resistor when **CHARGING** capacitor

Use **Kirchhoff's Loop Rule** to create the differential equations that describe RC circuits!

11.8.3

Steady State Conditions for Various Open/Closed Switch Configurations

Steady State

Steady State ≠ Equilibrium

BUT

~~Steady State ≈ Equilibrium~~

Our Textbook's Definition

Condition of a System as t->

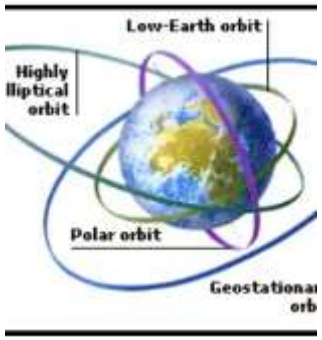
Generally Considered as:

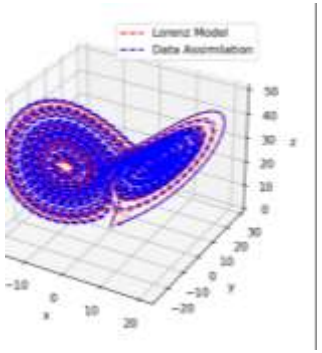
NOT

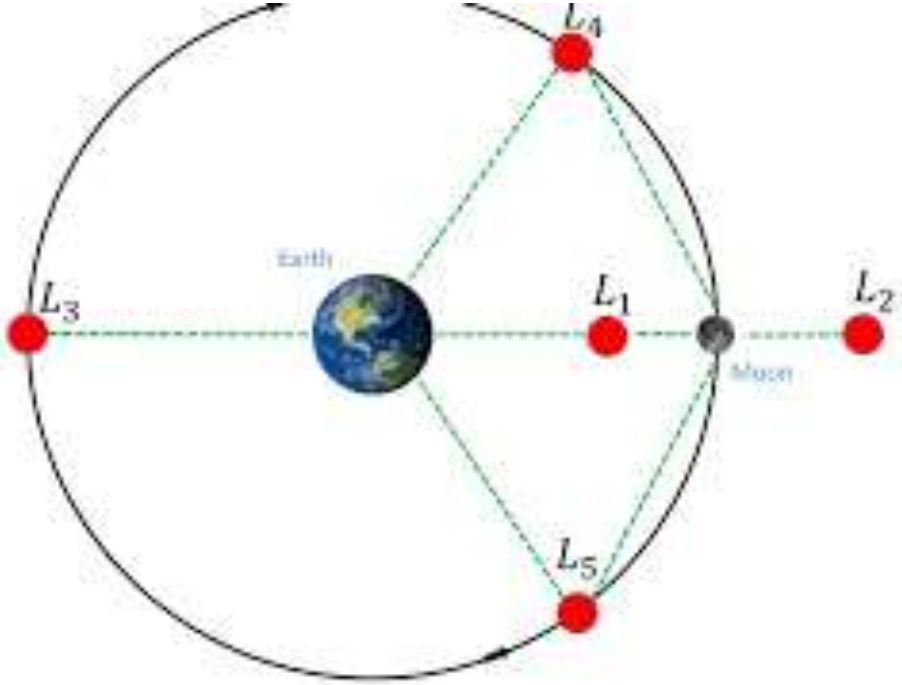
"Equilibrium Points/ Positions/ Values"

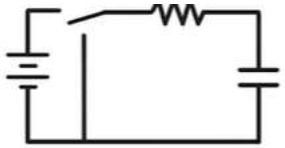
But Rather

Values at which the system will closely oscillate around









$$V_c(t) = V_b\left(1 - e^{-t/RC}\right)$$
Voltage across a **CHARGING** capacitor

$$V_c(t) = V_b e^{-t/RC}$$
Voltage across a **DISCHARGING** capacitor

$$Q(t) = CV_b\left(1 - e^{-t/RC}\right)$$
Charge on a **CHARGING** capacitor

$$Q(t) = Q_0 e^{-t/RC}$$
Charge on a **DISCHARGING** capacitor

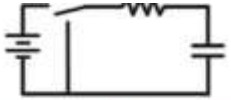
$$V_R(t) = V_b e^{-t/RC}$$
Voltage across the resistor when **CHARGING** capacitor

$$V_R(t) = V_b e^{-t/RC}$$
Voltage across the resistor when **DISCHARGING** capacitor

$$I(t) = \frac{V_b}{R} e^{-t/RC}$$
Current through resistor when **CHARGING** capacitor

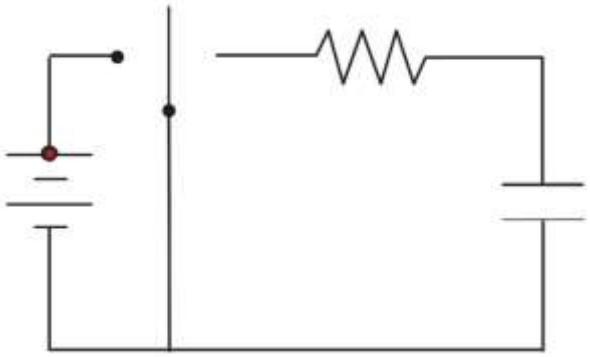
$$I(t) = -\frac{V_b}{R} e^{-t/RC}$$
Current through resistor when **DISCHARGING** capacitor

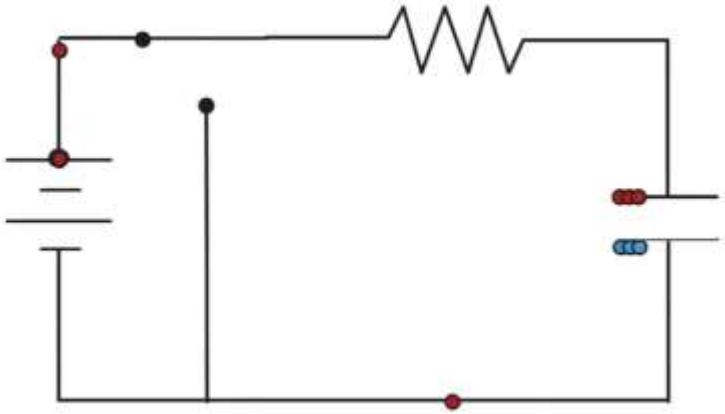
What happens to these equations as time goes to infinity?

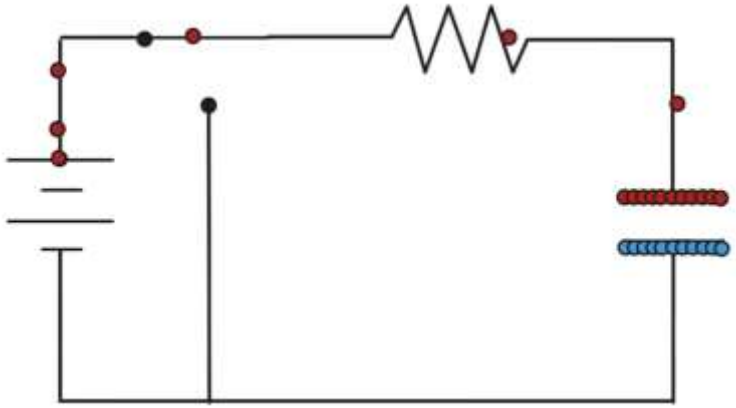


$V_c(t) = V_b(1 - e^{-t/RC})$ Voltage across a **CHARGING** capacitor	$V_c(t) = V_b e^{-t/RC}$ Voltage across a **DISCHARGING** capacitor
$Q(t) = CV_b(1 - e^{-t/RC})$ Charge on a **CHARGING** capacitor	$Q(t) = Q_0 e^{-t/RC}$ Charge on a **DISCHARGING** capacitor
$V_R(t) = V_b e^{-t/RC}$ Voltage across the resistor when **CHARGING** capacitor	$V_R(t) = V_b e^{-t/RC}$ Voltage across the resistor when **DISCHARGING** capacitor
$I(t) = \frac{V_b}{R} e^{-t/RC}$ Current through resistor when **CHARGING** capacitor	$I(t) = -\frac{V_b}{R} e^{-t/RC}$ Current through resistor when **DISCHARGING** capacitor

$V_c = V_b$ Voltage across a **fully charged** capacitor	$V_c = 0$ Voltage across an **EMPTY** capacitor
$Q = CV_b$ Charge on a **fully charged** capacitor	$Q = 0$ Charge on an **EMPTY** capacitor
$V_R = 0$ Voltage across the resistor when **fully charged** capacitor	$V_R(t) = V_b e^{-t/RC}$ Voltage across the resistor when **DISCHARGING** capacitor
$I = 0$ Current through resistor when **fully charged** capacitor	$I(t) = -\frac{V_b}{R} e^{-t/RC}$ Current through resistor when **DISCHARGING** capacitor



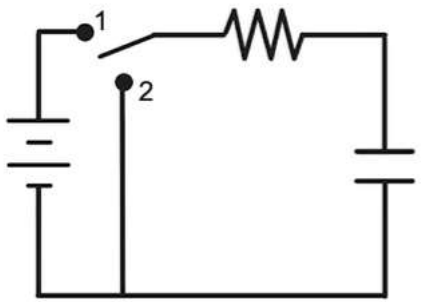




No more charge can be added to the capacitor, so no more current can pass through.
R->infinity,

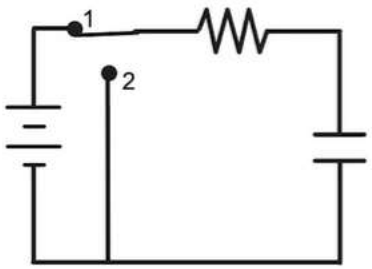
Capacitor initially uncharged

- $V_{battery} = 10\ V$
- $R = 100\ \Omega$
- $C = 100\ \mu F$



Capacitor initially uncharged

- $V_{battery} = 10\ V$
- $R = 100\ \Omega$
- $C = 100\ \mu F$



Move switch to position 1!

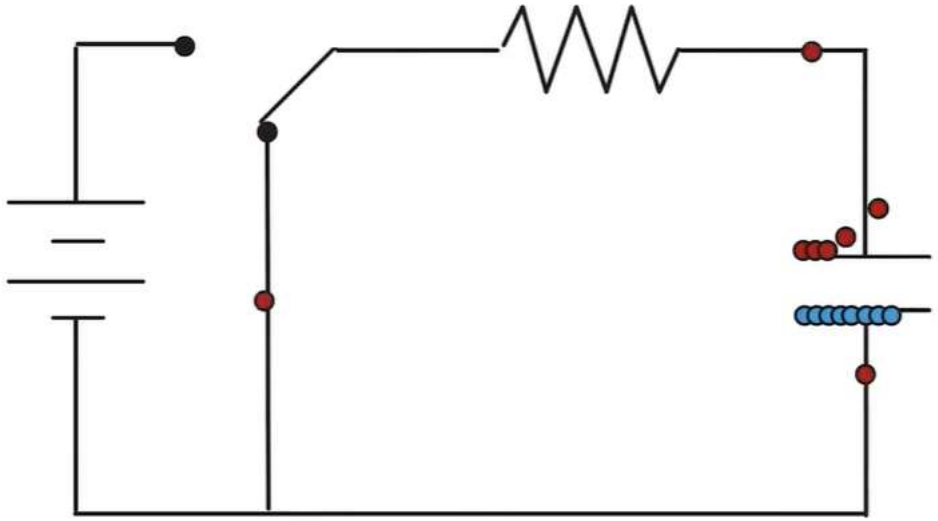
- RIGHT after you close the switch
- Capacitor is totally empty, acts like wire
  - $V = 10V$ across resistor
  - $I = \frac{10V}{100\Omega} = 0.1A$ through R

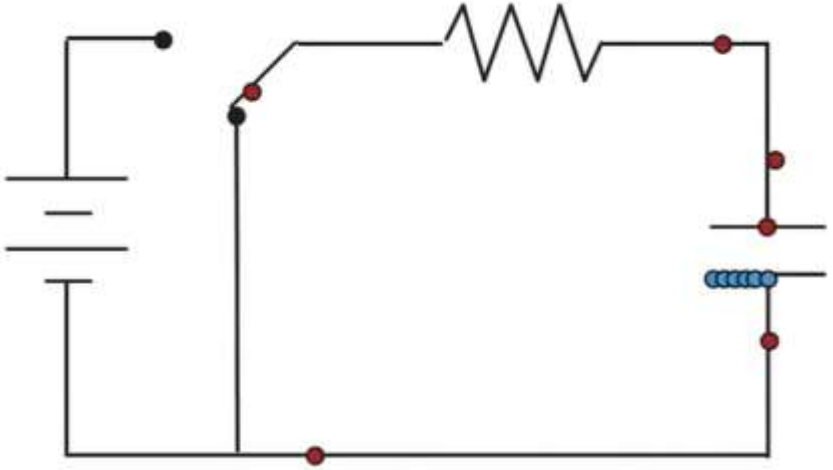
- After a long time, capacitor is full
- No current through resistor
  - $V = 0V$ across resistor
  - $V = 10V$ across capacitor

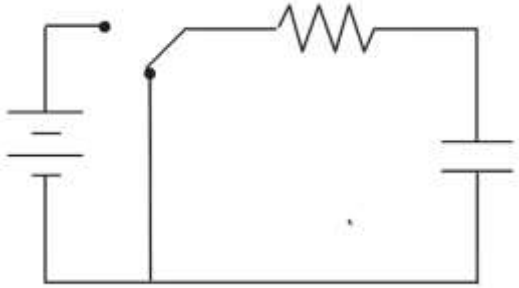
$$Q = CV$$
$$Q = (100 \cdot 10^{-6}F)(10V)$$
$$Q = 1\ mC$$

Move to Position 2

-Battery Removed
-Capacitor takes over
- “Discharging Capacitor”

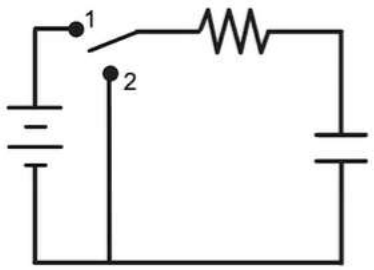






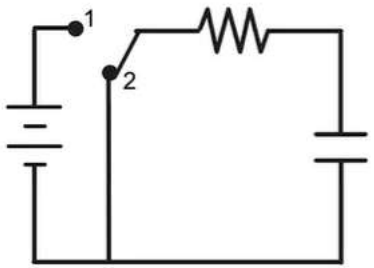
Capacitor is now fully charged

- $V_{battery} = 10\,V$
- $R = 100\,\Omega$
- $C = 10\,\mu F$



Capacitor is now fully charged

- $V_{battery} = 10\ V$
- $R = 100\ \Omega$
- $C = 10\ \mu F$

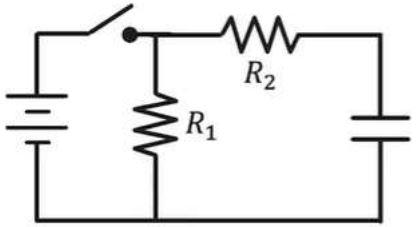


Move switch to position 2!

- RIGHT after you close the switch
  - $V = 10V$ across resistor
  - $I = \frac{10V}{100\Omega} = 0.1A$ through resistor

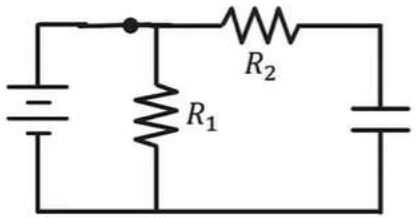
- After a long time, capacitor is empty
- No current through resistor
  - $V = 0V$ across resistor
  - $V = 0V$ across capacitor

Capacitor initially uncharged



- $V_{battery} = 10\ V$
- $R_1 = 100\ \Omega$
- $R_2 = 200\ \Omega$
- $C = 100\ \mu F$

Capacitor initially uncharged



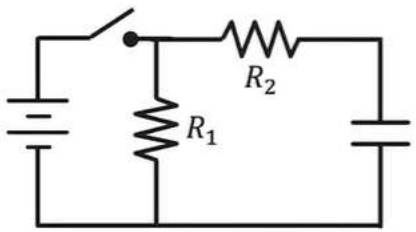
- $V_{battery} = 10\ V$
- $R_1 = 100\ \Omega$
- $R_2 = 200\ \Omega$
- $C = 100\ \mu F$

Close the switch!

- RIGHT after you close the switch
- Capacitor is totally empty, acts like wire
  - $V = 10V$ across each resistor
  - $I = \frac{10V}{100\Omega} = 0.1A$ through R_1
  - $I = \frac{10V}{200\Omega} = 0.05A$ through R_2

- After a long time, capacitor is full
- No current through R_2
  - $V = 0V$ across R_2
  - $V = 10V$ across capacitor and R_1
  - $I = \frac{10V}{100\Omega} = 0.1A$ through R_1
  - $Q = (100\mu F)(10V) = 1mC$ full charge

Capacitor initially charged



- $V_{battery} = 10\ V$
- $R_1 = 100\ \Omega$
- $R_2 = 200\ \Omega$
- $C = 100\ \mu F$

Open the switch!

- RIGHT after you open the switch
- Capacitor is totally full, acts like battery
  - $V = 10V$ across resistors in SERIES
  - $I = \frac{10V}{300\Omega} = 0.033\ A$ through R_1 and R_2
- After a long time, capacitor is empty
- No current through resistors
  - $V = 0V$ across both resistor
  - $V = 0V$ across capacitor

Takeaways

- **Uncharged capacitors** initially act as wires until charge builds up.
- **Fully charged** capacitors act as roadblocks.

11.8.4

Non-Steady State Conditions for Various Open/Closed Switch Configurations

$$V_c(t) = V_b\left(1 - e^{-t/RC}\right)$$
Voltage across a **CHARGING** capacitor

$$V_c(t) = V_b e^{-t/RC}$$
Voltage across a **DISCHARGING** capacitor

$$Q(t) = CV_b\left(1 - e^{-t/RC}\right)$$
Charge on a **CHARGING** capacitor

$$Q(t) = Q_0 e^{-t/RC}$$
Charge on a **DISCHARGING** capacitor

$$V_R(t) = V_b e^{-t/RC}$$
Voltage across the resistor when **CHARGING** capacitor

$$V_R(t) = V_b e^{-t/RC}$$
Voltage across the resistor when **DISCHARGING** capacitor

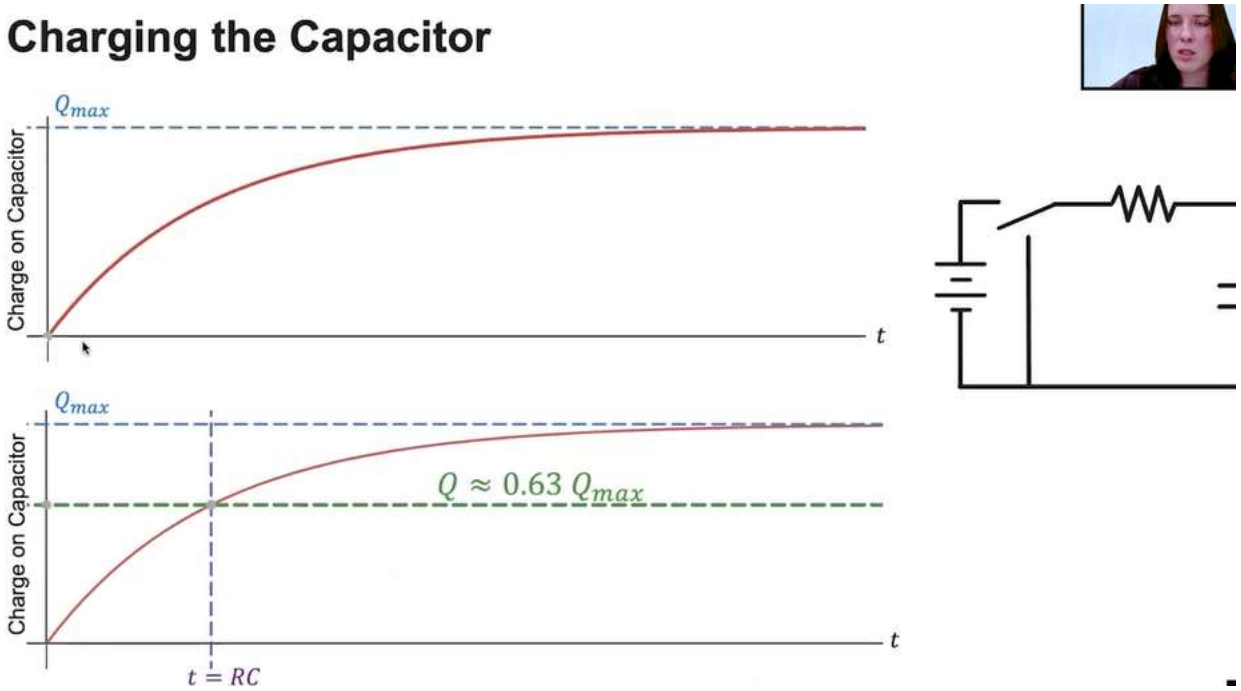
$$\mathrm{I}(t) = \frac{V_b}{R} e^{-t/RC}$$
Current through resistor when **CHARGING** capacitor

$$\mathrm{I}(t) = -\frac{V_b}{R} e^{-t/RC}$$
Current through resistor when **DISCHARGING** capacitor

RC time constant

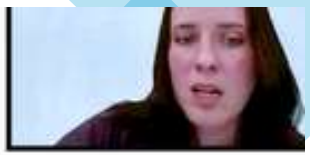
The time constant of an RC circuit is a measure of how quickly the capacitor will charge or discharge.

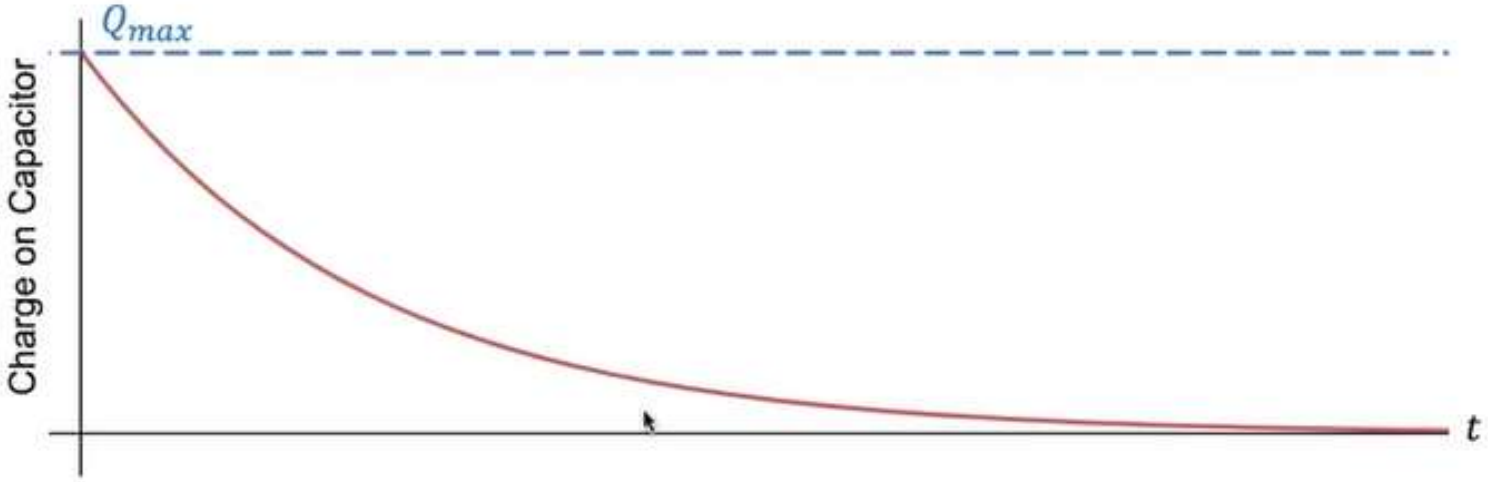
$$\tau = R_{eq} C_{eq}$$

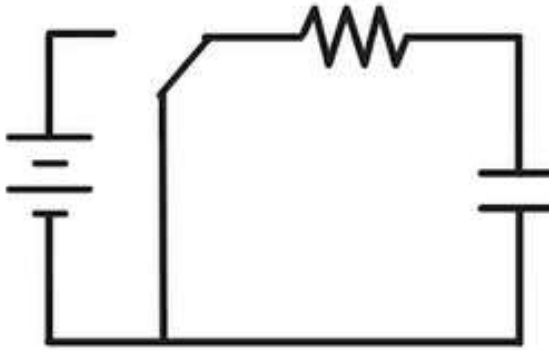


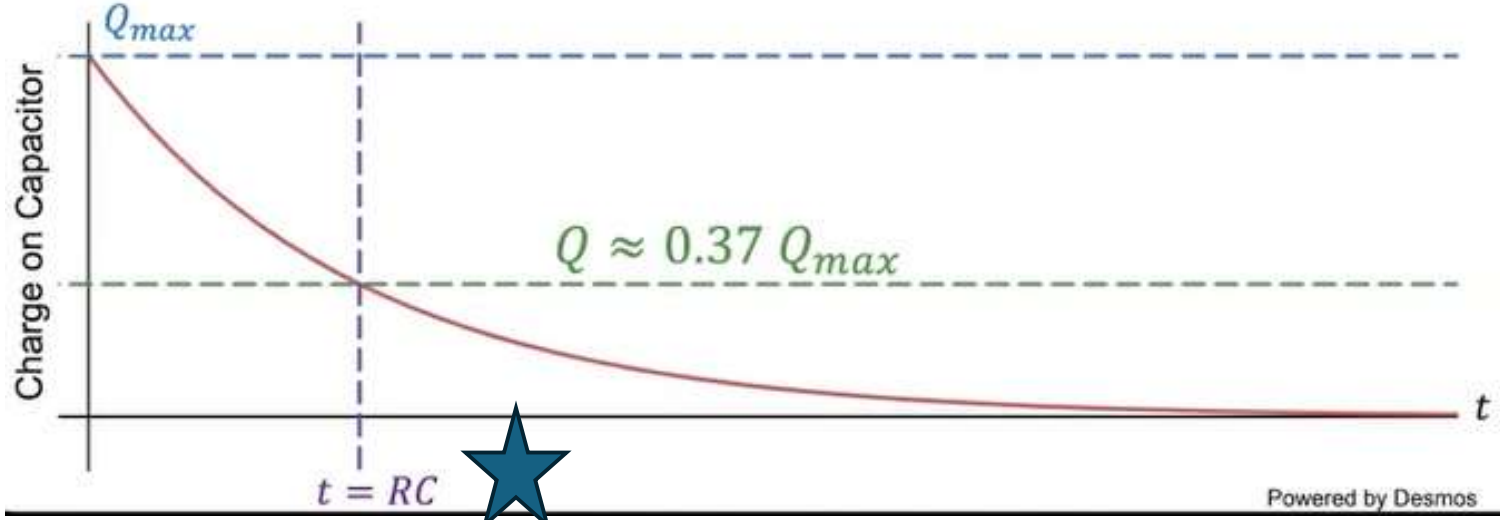
-According to the video, you “MAY” be ask
“When will an RC capacitor reach 63% max charge?”

Discharging the Capacitor









-According to the video, you "MAY" be ask
"When will an RC capacitor discharge to 37% max charge?"

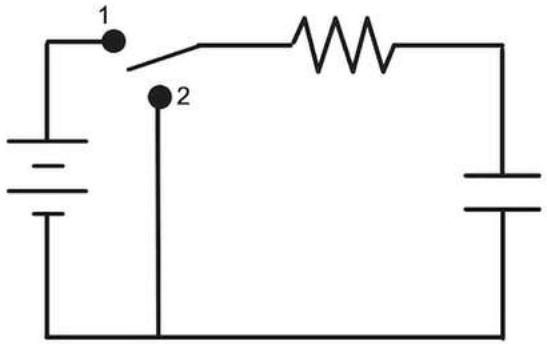
But how do I know which equation to use?

$$e^{-t/RC} \quad \text{vs.} \quad 1 - e^{-t/RC}?$$

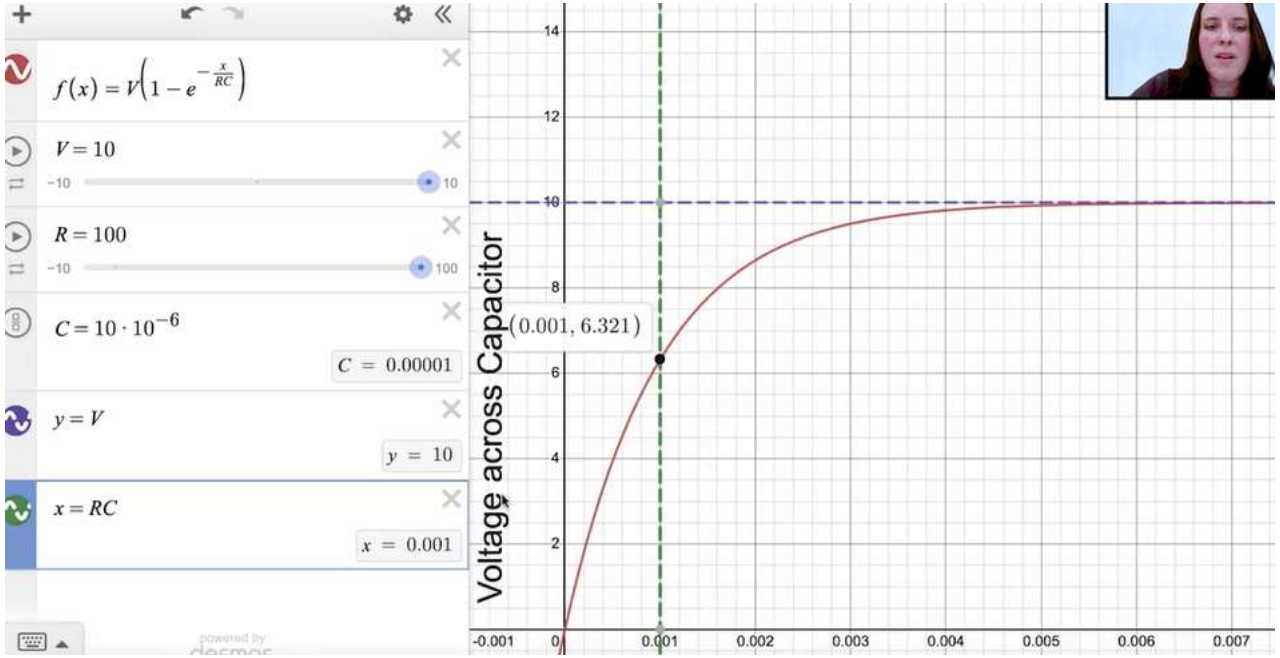
Decay= Discharging

Growth= Charging

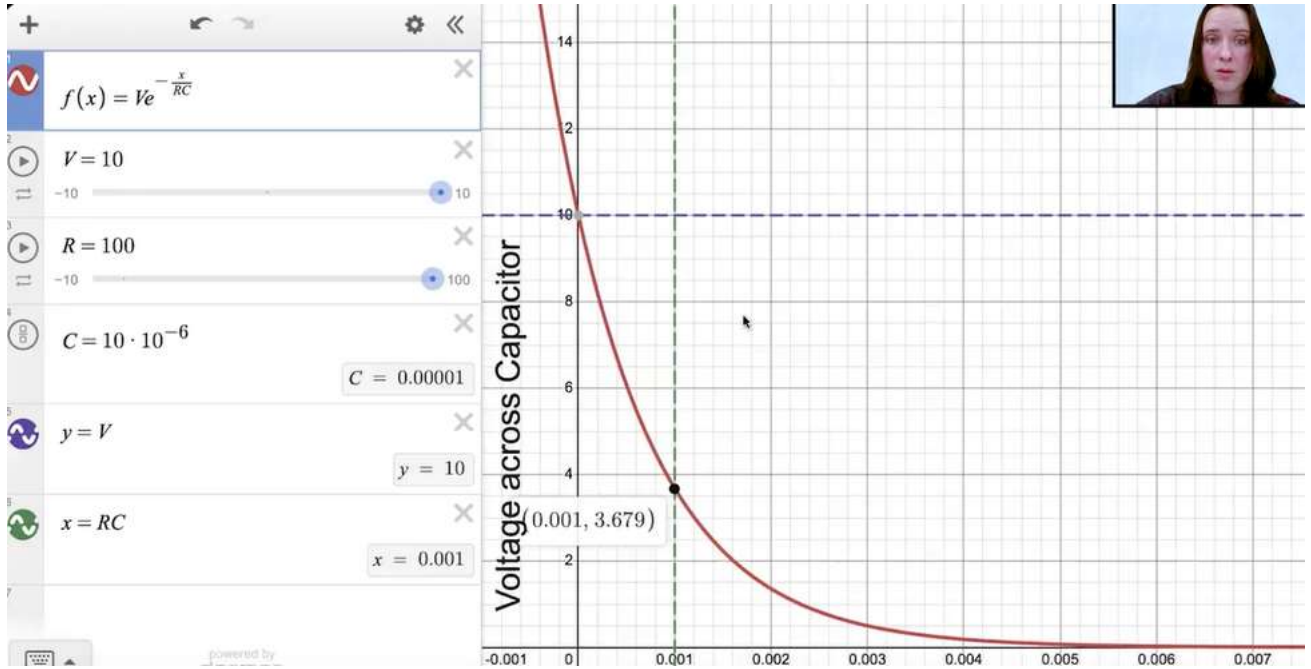
- Capacitor initially **uncharged**
- $V_{battery} = 10\ V$
- $R = 100\ \Omega$
- $C = 10\ \mu F$
- Flip switch to **position 1**!

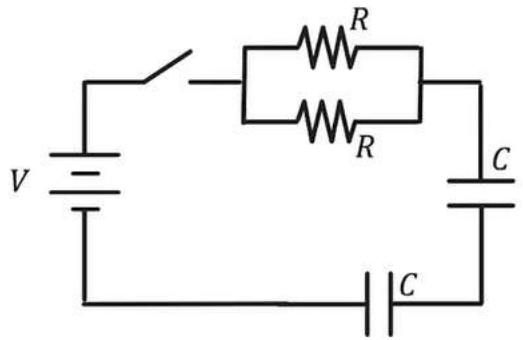


Charging

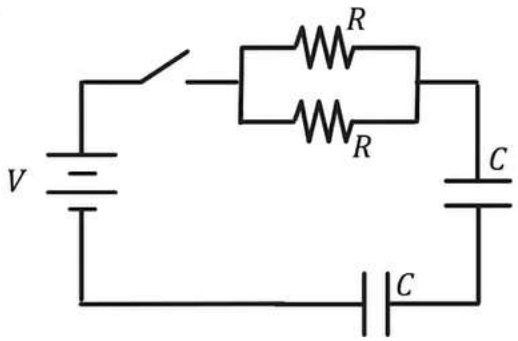

$f(x) = V\left(1 - e^{-\frac{x}{RC}}\right)$
$V = 10$
$R = 100$
$C = 10 \cdot 10^{-6}$
$C = 0.00001$
$y = V$
$y = 10$
$x = RC$
$x = 0.001$
(0.001, 6.321)
Voltage across Capacitor

Discharging





When will the capacitors reach 63% of their max charge?
When discharging, when will the capacitors reach 37% of their max charge?



$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$$

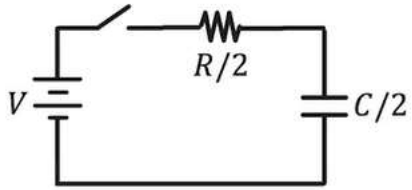
$$R_{eq} = \frac{R}{2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{C} = \frac{2}{C}$$

$$C_{eq} = \frac{C}{2}$$

$$\tau = R_{eq}C_{eq} = \frac{R}{2} \cdot \frac{C}{2}$$

$$\tau = \frac{RC}{4}$$



When will the capacitors reach 63% of their max charge?
When discharging, when will the capacitors reach 37% of their max charge?

Takeaways

Think about whether the quantity in question is a growth or a decay

$$e^{-t/RC} \quad \text{vs.} \quad 1 - e^{-t/RC}$$